The Eyes Don't Lie -- Predicting Your Decisions Before They're Made

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Introduction

- Eye movement parameters (e.g., total looking time) reflect a number of cognitive processes including decision making (e.g., Krajbich et al. 2010 -- drift diffusion model)
- The goals of the current study are:
  a. To extend previous research and examine when eye data can reliably predict choice.
  b. To determine if different levels of decision making affects eye movement parameters.
  c. To explore if looking time could be used as a temporal index for decision awareness.

- We monitored eye movements while subjects performed a visual decision task: deciding which of two novel black/white objects was more symmetrical (objective judgment) or more beautiful (subjective judgment).

- Participants spent more time looking at the eventually chosen item.

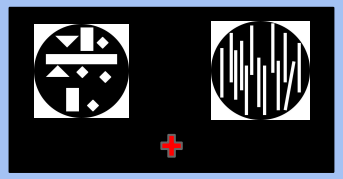
- Results from time series analysis suggests that the moment of this difference in looking time may serve as a temporal marker for awareness of visual decision.

Methods

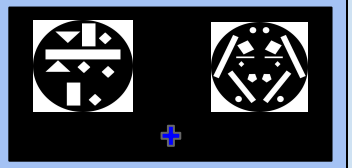
- We made stimuli pairs to create two trial types:
Preference (red, matching stimuli types) or
Symmetry (blue, symmetrical paired with asymmetrical).

IVs: Objectivity level of decision (Symmetry/Preference), time before decision (500 ms bins)

DVs: Total looking time, first and last fixations, number of saccades (eye movements between areas)



Preference Trial
Left stimulus is asymmetrical
Right stimulus is asymmetrical



Symmetry Trial
Left stimulus is asymmetrical
Right stimulus is symmetrical

Procedure

- We asked subjects to choose which of the two stimuli they found more beautiful or more symmetrical.

- Each stimulus was uniquely and randomly paired with 7 other stimuli.

- This created a set of **91 trials** (43 symmetry pairings and 48 preference pairings).

- While subjects were viewing the objects, we monitored their eye movements (Eyelink, sampled at 500Hz).

Acknowledgements

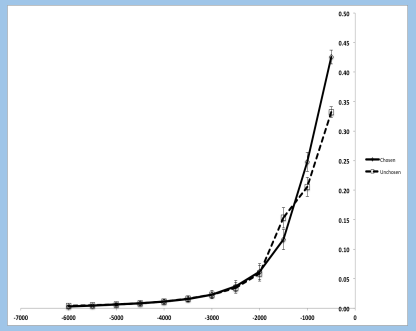
- Nick DiQuatro and Rachel Gwinn

Results

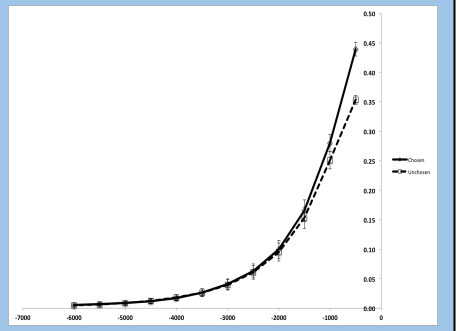
- The proportional looking time spent on the chosen item was longer than on the unchosen item, $t(19) = 9.47$, $p < 0.001$ for both types of decisions.

- Time series analysis results show this difference in looking time to be reliable at **1000 ms** and **1500 ms** prior to symmetry and preference decisions, respectively, $t(19) > 3.8$, $p < 0.001$.

- Reaction times: Symmetry 2452 ms (SE = 200), Preference 2986 ms (SE = 212), $t(19) = 5.11$, $p < 0.0001$



Preference Decision



Symmetry Decision

Discussion

- Cumulative looking time predicts choice regardless of the objectivity level of the choice, making it applicable to any visual decision task.

- However, it takes participants longer to explicitly state a subjective choice, as indicated by the time series analysis.

- In progress: investigate whether cumulative looking time and the 'moment of decision' can be applied to consciousness research.

- To do so, we need to couple the eye data with awareness report (e.g., Libet's moment of intent "W").

- This future experiment will hopefully allow us to better quantify the concepts of consciousness and awareness in decision making by examining when a subject truly makes his/her decision.